KIET DEEMED TO BE UNIVERSITY

SOFTWARE ENGINEERING LABORATORY

LAB EXPERIMENT - 2

Use Case Diagram and Use Case Specifications

SMART WASTE COLLECTION MANAGEMENT SYSTEM

Submitted by: GROUP VIII
Avni Khare - 202401100200105
Awanish Chaudhary - 202401100200106
Ayush Chaurasia - 202401100200107
Ayush Kumar Yadav - 202401100200108

Based on IEEE 830-1998 Software Requirements Specification

1. Aim

To draw the UML use case diagram for the Smart Waste Collection Management System (SWCMS), identify the roles of its actors, and specify the preconditions, functions and postconditions of each use case.

2. Actors and Their Roles

Actor	Role
Administrator	Manages users, waste bins, collection tasks, complaints, alerts and reports.
Waste Collector	Views assigned tasks, carries out waste collection and updates task status.
Citizen	Submits waste-related complaints and tracks their status.

Optional external actor: An IoT sensor or external bin data source may supply fill-level readings in a future version. Real-time IoT integration is not required for the initial implementation.

3. Use Case Diagram

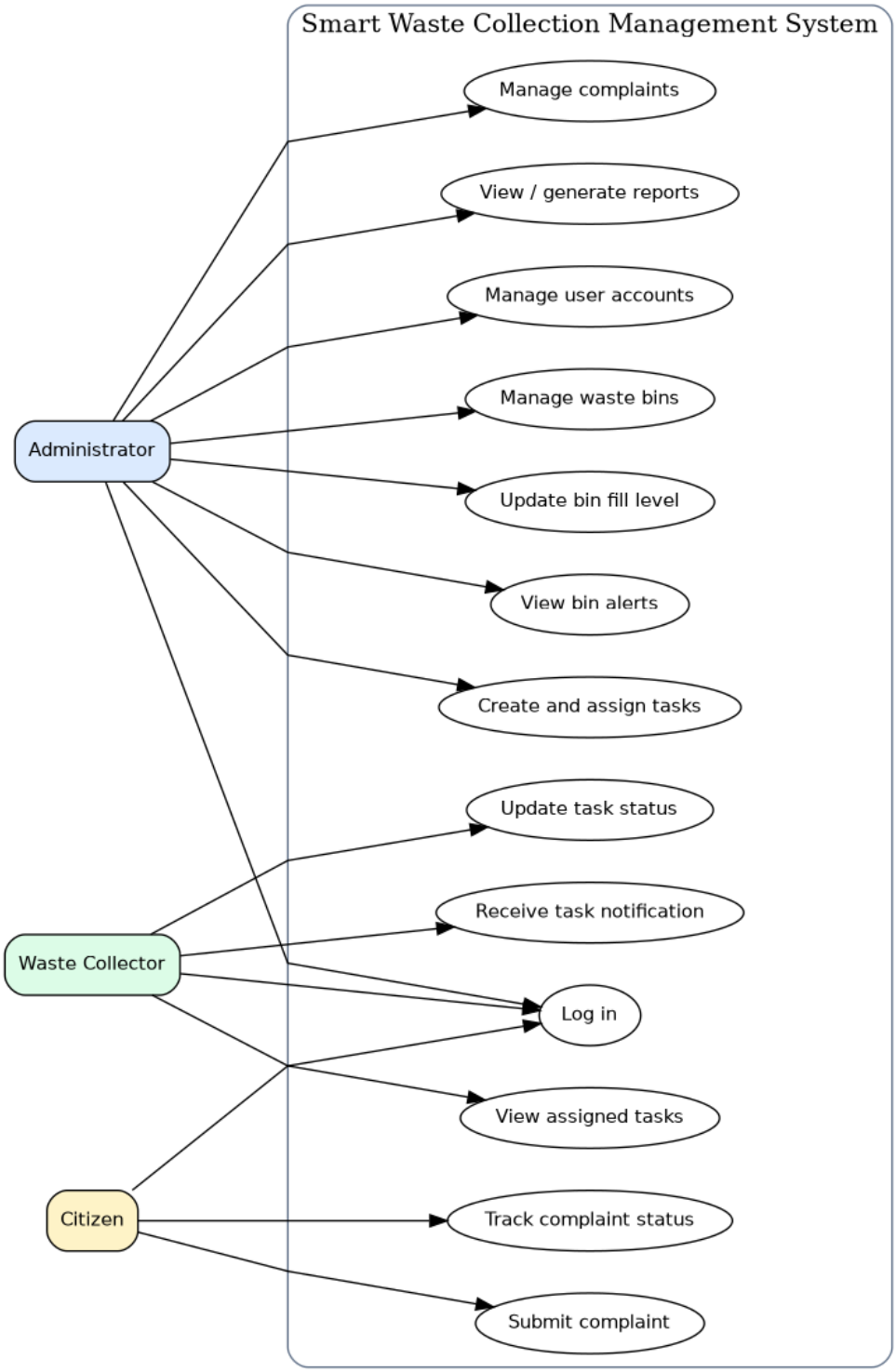


Figure 1. Use case diagram of the SWCMS.

4. Use Case Specifications

Precondition: What must be true before a use case begins. Postcondition: What becomes true after successful completion.

UC-01: User Login

Actor: Administrator, Waste Collector, Citizen
Precondition: User has a valid registered account.
Function: Authenticates credentials and identifies the user role.
Postcondition: User is authenticated and can access the appropriate dashboard.

UC-02: Manage User Accounts

Actor: Administrator
Precondition: Administrator is logged in and user management is enabled.
Function: Adds, updates and manages accounts and assigned roles.
Postcondition: Account information is saved in the database.

UC-03: Manage Waste Bins

Actor: Administrator
Precondition: Administrator is logged in.
Function: Adds bins, edits location and bin details, and deactivates obsolete bins.
Postcondition: Bin records are updated in the database.

UC-04: Update Bin Fill Level

Actor: Administrator (manual update)
Precondition: The waste bin is registered.
Function: Records the bin fill level manually or through an integrated source.
Postcondition: Fill level is stored and a threshold alert is generated when applicable.

UC-05: View Bin Alerts

Actor: Administrator
Precondition: Administrator is logged in and bin status data is available.
Function: Shows bins that have reached a configured collection threshold.
Postcondition: Administrator can identify bins requiring attention.

UC-06: Create and Assign Collection Tasks

Actor: Administrator
Precondition: Administrator is logged in; bins and an available collector exist.
Function: Creates tasks, selects priority and schedule, and assigns a collector.
Postcondition: The task is stored and assigned to the collector.

UC-07: View Assigned Tasks

Actor: Waste Collector
Precondition: Collector is logged in and assigned tasks exist.
Function: Displays assigned tasks with bin or area, priority and schedule.
Postcondition: Collector can review work to be performed.

UC-08: Update Task Status

Actor: Waste Collector
Precondition: Collector is logged in and has an assigned task.
Function: Changes status to Pending, In Progress or Completed.
Postcondition: New status is saved; completion time is recorded when completed.

UC-09: Submit Complaint

Actor: Citizen
Precondition: Citizen has access to the complaint submission interface.
Function: Submits a waste-related issue category and description.
Postcondition: Complaint is stored with a unique identifier.

UC-10: Track Complaint Status

Actor: Citizen
Precondition: A complaint has been submitted and its status is available.
Function: Views the current status of the submitted complaint.
Postcondition: Citizen can see the latest recorded complaint status.

UC-11: Manage Complaints

Actor: Administrator
Precondition: Administrator is logged in and complaint records exist.
Function: Views complaints and updates their resolution status.
Postcondition: Complaint status and history are saved.

UC-12: Receive Task Notifications

Actor: Waste Collector
Precondition: A new task is assigned and notification functionality is available.
Function: Provides notification of newly assigned tasks.
Postcondition: Notification is available to the collector.

UC-13: Generate Reports

Actor: Administrator
Precondition: Administrator is logged in and relevant records are available.
Function: Displays bin summaries, task reports by status and date, and complaint summaries.
Postcondition: Operational reports are displayed on screen.

5. Use Case Relationships

Association: Connects an actor with the use cases they perform. Include: Task assignment may be represented as a required part of the combined create-and-assign workflow. Extend: A threshold alert is generated conditionally when an updated bin fill level reaches the configured threshold.

6. Result

The SWCMS use case diagram and specifications were prepared for the three primary actors. They describe user access, waste-bin monitoring, collection scheduling, complaint management, notifications and reporting.

Source: SWCMS Software Requirements Specification (IEEE 830-1998 style), Version 1.0, August 2026. Citizen complaint tracking is described in the SRS user characteristics; advanced IoT and route optimization remain future enhancements.